Exact Key-List Recovery in SEPAR

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Abstract

SEPAR's word permutations preserve nested blocks. State additions disturb these blocks only at narrow carry boundaries. We use this structure to recover numerical rotor keys from quotient paths, then remove rotors inward by an exact partition join. Initialization supplies the remaining state words and the first two rotor keys. The result is a chosen-IV algorithm that returns a list of complete 256-bit keys without ranking or candidate cutoffs. For every fixed master key, its true key is in the list with probability at least $1 - 1.307164 \cdot 10^{-8}$. We also give a 511-query distinguisher.

1 Target and attack model

We analyze the literal supplied C implementation of SEPAR [1]. It has eight keyed word permutations, a 128-bit state, and a 16-bit LFSR. The published description and the C source differ in some sequential assignments; the equations here follow the source. An adversary may choose an IV and a plaintext, reset, and ask for the first ciphertext word repeatedly under that IV. This reuse is part of the attack model. The recovery never reads the internal state or a rotor oracle. Complete candidate keys are also checked against saved public ciphertext tables.

All additions below are modulo 2^{16}. Write a word as four nibbles (a, b, c, d), from most to least significant. Let Σ apply the four S-boxes in parallel. Their entries for inputs $0, \ldots, F$ are

S_1	1	F	B	2	0	3	5	8	6	9	C	7	D	A	E	4
S_2	6	A	F	4	E	D	9	2	1	7	C	B	0	3	5	8
S_3	C	2	6	1	0	3	5	8	7	9	B	E	A	D	F	4
S_4	D	B	2	7	0	3	5	8	6	C	F	1	A	4	9	E

Rotor i has its own two-word key $K_i = (k_{i,0}, k_{i,1})$. Its four subkeys are

$$\begin{aligned} G(u) &= u \text{ OR } (S_1((u \gg 6) \,\&\, 15) \ll 6), \\ \kappa_0 &= k_{i,0}, \qquad \kappa_1 = k_{i,1}, \\ \kappa_2 &= G(\mathrm{rol}_{16}(k_{i,0}, 6)) \oplus (i+2), \\ \kappa_3 &= G(\mathrm{rol}_{16}(k_{i,1}, 10)) \oplus (i+3). \end{aligned}$$

For $j = 0, 1, 2, 3$, apply $x \leftarrow L\Sigma(x \oplus \kappa_j)$; finally apply $x \leftarrow \Sigma(x \oplus \kappa_0 \oplus \kappa_1) \oplus \kappa_2 \oplus \kappa_3$. The implemented linear map is

$$L(a, b, c, d) = (b \oplus c, b, a, a \oplus d). \tag{1}$$

This defines a permutation $\mathcal{E}_i$, with inverse $\mathcal{D}_i$. The OR in G is intentional: replacing bits instead would describe a different target.

For an initialized state $s = (s_1, \ldots, s_8)$, the first-word map is

$$R_0(p) = p, \qquad R_i(p) = \mathcal{E}_i(R_{i-1}(p) + s_i), \qquad C(p) = R_8(p). \tag{2}$$

Initialization consists of four rounds of the map given in Section 2. Later words use a state and LFSR update specified by the source. All acquisition queries in this attack reset before the first word, so (2) is the map under analysis. Complete-key replay uses the source's initialization.

2 Preserved blocks and useful states

Lemma 2.1 (Triangular rotor)**.** *Every numerical rotor key gives*

$$\mathcal{E}_i(a,b,c,d) = \big(P_{i,b}(a),\, Q_i(b),\, M_{i,b}(c),\, T_{i,a,b,c}(d)\big),$$

where each one-coordinate map is a permutation. Consequently, the high twelve output bits depend only on the high twelve input bits. The middle-byte quotient

$$m_i(16b+c) = 16Q_i(b) + M_{i,b}(c)$$

is a permutation of 256 values.

Proof. One keyed substitution and linear layer send (a,b,c,d) to coordinates depending respectively on $(b,c), b, a, (a,d)$. Four layers give dependency sets $(a,b), b, (b,c), (a,b,c,d)$. The final substitution and XOR do not enlarge them. With the other coordinates fixed, each coordinate path is a composition of nibble permutations and XORs. □

In particular, a change confined to d stays in d under L; its nibble branch number is two. A carry at a state addition may move an input between two adjacent quotient blocks, but cannot diffuse it through all blocks. These narrow carry boundaries are the objects decoded below. The OR schedule also leaves small raw-key fibres after quotient data have been fixed.

One initialization round first computes

$$v_1 = \mathcal{E}_1(s_1+s_3+s_5+s_7), \qquad v_i = \mathcal{E}_i(v_{i-1}+s_i) \quad (2 \le i \le 8),$$

then replaces s_1 by s_1+v_8 and s_i by s_i+v_{i-1} for $i \ge 2$.

Lemma 2.2 (Uniform state)**.** *For every fixed key, four initialization rounds permute the 128-bit IV space. Independent uniform IVs therefore give independent uniform initialized state words.*

Proof. Given the output t of one round, invert in the order

$$s_1 = t_1 - \mathcal{E}_8(t_8), \quad s_i = t_i - \mathcal{E}_{i-1}(t_{i-1})\ (i = 8,\dots,3), \quad s_2 = t_2 - \mathcal{E}_1(s_1+s_3+s_5+s_7).$$

Four bijections compose to a bijection. □

Let a block have B consecutive words. An input row consists of all its B words. The next lemma is the same induction at every quotient width used by the attack.

Lemma 2.3 (Perturbed blocks)**.** *Align a row with the first state addition. Write the intervening state additions before a chosen last rotor as $Bh_i + e_i$, with signed residues e_i, and put $S = \sum_i |e_i|$. If the final state residue is b, with*

$$S < \min(b, B-b), \tag{3}$$

the two endpoints of the last quotient-cycle edge both occur in the row, with at least $B-S$ points between them. At least $B-S$ inverse images under the last rotor lie in one interval $[Bv+b, Bv+b+B-1]$. If $2S < \min(b, B-b)$, the endpoints are the unique two largest quotient-label counts.

Proof. After the first rotor, an aligned row fills one block. At the next translation, at most $|e_i|$ distinct low coordinates cross a boundary. Discard them and apply the rotor; it permutes the remaining coordinates within a nominal block. Induction leaves at least $B-S$ points, and the nominal block labels are a permutation of their original labels. The final addition by b splits the full nominal block into parts of sizes $b, B-b$. Condition (3) leaves each part nonempty after the discarded points are removed. All retained points lie in the stated translated interval before the final rotor. If $2S$ is below each part size, each intended label has more points than all the other labels combined, proving dominance. □

3 A preliminary distinguisher

The block weakness already distinguishes SEPAR without numerical key recovery.

Theorem 3.1 (511-query distinguisher)**.** *At one IV, query $p = 0, \dots, 510$, resetting before each query. Accept if some consecutive 256-input window has at most 128 ciphertext high-byte values. Every SEPAR key and IV is accepted. A uniform random word permutation is accepted with probability at most* $1.762252054731 \cdot 10^{-9} < 2^{-29}$.

Proof. One of the 256 windows aligns $p + s_1$ with a complete high-byte block. Equation (1) makes the first rotor's high byte constant on it. Each of the seven later additions can split a high-byte block into at most two, while a rotor merely permutes the classes. Thus the ciphertext uses at most 2^7 high bytes.

For the ideal permutation, let $A_{r,k}$ count ordered selections of r distinct words occupying k high-byte classes. With $A_{0,0} = 1$ and infeasible entries zero,

$$A_{r+1,k} = A_{r,k}(256k - r) + 256(257 - k)A_{r,k-1}.$$

The two terms reuse a class or open a new one. Write $(a)_b = a(a-1)\cdots(a-b+1)$. The probability for one window is

$$\frac{\sum_{k \le 128} A_{256,k}}{(65536)_{256}}.$$

A union bound over the 256 windows and exact integer evaluation of this recurrence give the stated value. □

We now turn to complete key recovery.

4 Decoding a rotor from middle paths

This section supplies a numerical decoder shared by outer exposure and the inward factorization. For a rotor subkey write $\kappa_j = (\alpha_j, \beta_j, \gamma_j, \delta_j)$. Define

$$J^{u,v,w,t}(x) = S_3(S_1(S_3(S_1(S_3(x) \oplus u) \oplus v) \oplus w) \oplus t)\,.$$

Substitution into the four literal rounds yields

$$\begin{aligned} M_b(c) &= (\gamma_2 \oplus \gamma_3) \oplus J^{u_b,v,w_b,t}(c \oplus \gamma_0), \\ u_b &= S_2(b \oplus \beta_0) \oplus \alpha_1, \quad v = \gamma_2, \quad t = \gamma_0 \oplus \gamma_1, \\ w_b &= S_2(S_2(S_2(b \oplus \beta_0) \oplus \beta_1) \oplus \beta_2) \oplus \alpha_3. \end{aligned} \tag{4}$$

Thus, for each output lane $y = Q(b) + \lambda \bmod 16$, the sequence $M_b(0), \dots, M_b(15)$ is a path through its sixteen vertices. Its orientation and the common cyclic lane translation λ may be unknown.

For a nibble table f, put

$$\operatorname{pack}(f) = \sum_{x=0}^{15} f(x) 2^{4x}, \qquad \mathcal{C}(f) = \min_{a=0}^{15} \operatorname{pack}\big(x \mapsto f(x \oplus a) \oplus f(a)\big).$$

This canonicalizes input and output XOR. Reversing a path means $c \mapsto 15 - c = c \oplus 15$, so it does not change the canonical value. Let $I = S_2^{-1}$ and

$$F_{j,r,s}(y) = I(I(I(I(y \oplus r \oplus s) \oplus j) \oplus s) \oplus r). \tag{5}$$

Two finite identities, computed from the S-boxes printed above, are used throughout:

Family	Tables	Distinct signatures
$\mathcal{C}(J^{u,v,w,t})$	65536	65536
$\mathrm{pack}(y \mapsto F_{j,r,s}(y-\lambda) \oplus F_{j,r,s}(-\lambda))$	65536	65536

The first calculation loops over all four nibbles (u, v, w, t), evaluates all sixteen entries, and sorts their canonical 64-bit words. The second loops over (j, r, s, λ), evaluates (5) on all sixteen inputs, and sorts those packed words. Equality of adjacent sorted entries is the complete collision test. Seven first-family tables have a nonidentity input/output XOR stabilizer. These are exhaustive finite assertions, not estimates over master keys.

Theorem 4.1 (Forty numerical keys)**.** *Sixteen supplied unoriented middle paths, indexed by output lane up to a common cyclic translation, determine at most forty raw 32-bit rotor keys. The exact decoder emits all compatible keys and the compatible translation. It includes the true key for true paths.*

Proof. Canonicalize each path. The first finite identity determines its (u_y, v_y, w_y, t_y); reject if the v_y, t_y disagree. The u_y must be distinct. Set $j = \beta_0 \oplus \beta_1$, $r = \beta_2$, $s = \beta_3$, and $\kappa = \beta_1 \oplus \alpha_1$. Inverting the autonomous output lane gives

$$u_y = F_{j,r,s}(y - \lambda) \oplus \kappa.$$

The second finite identity applied to $u_y \oplus u_0$ fixes (j, r, s, λ); u_0 fixes κ. Then

$$\alpha_3 = w_y \oplus S_2(S_2(u_y \oplus \kappa) \oplus r)$$

must be the same in every lane.

At least one path core has no nonidentity stabilizer: there are sixteen distinct u_y and only seven exceptional cores in the entire family. This path fixes γ_0 up to reversal; the key schedule fixes $\gamma_0 \bmod 4 = r \gg 2$, choosing one orientation. Equation (4) then fixes γ_3. The implementation checks the equations on every row, so it does not select a special row or key value.

Write $g(x) = x \,\mathrm{OR}\, S_1(x)$. The schedule gives

$$\beta_1 \gg 2 = \gamma_3 \,\&\, 3, \quad g(\alpha_1) = (\gamma_3 \gg 2) + 4(s \,\&\, 3), \quad \alpha_1 = \beta_1 \oplus \kappa.$$

For each fixed upper two bits, every fibre of g has size at most two, as is checked directly from

$$(g(0), \ldots, g(15)) = (1, 15, 11, 3, 4, 7, 7, 15, 14, 9, 14, 15, 13, 15, 14, 15).$$

There are therefore at most two β_1, four lower-two-bit choices for α_0, and five choices for δ_0 after its g-value is fixed. The remaining raw-key nibbles follow from the displayed schedule. Checking the complete paths leaves at most $2 \cdot 4 \cdot 5 = 40$ keys, all of which are emitted. □

For a complete quotient cycle let $q_K(x) = \mathcal{E}_K(16x) \gg 4$. A translated cycle has order

$$q_K(\varepsilon x + a) + h \pmod{4096}, \qquad \varepsilon \in \{1, -1\}, \tag{6}$$

up to its cyclic starting point. Try $h \bmod 16$, reduce the result modulo 256, and extract its sixteen middle paths. Theorem 4.1 fixes $h/16 \bmod 16$. For each emitted key, try the sixteen remaining top-nibble translations and compare all 4096 entries in both directions. This finite procedure emits every compatible numerical rotor and full quotient translation; it has no ranking cutoff.

5 Recovering the outer rotor

We use every pair of quotient labels supported by a row. This avoids a dominance assumption: false edges may enter the graph, but the true path remains.

Align a row of sixteen plaintexts at $r = -s_1 \bmod 16$. For $i = 2, \ldots, 7$, write $s_i = 16h_i + e_i$, using signed residues, and let $S = \sum_{i=2}^{7} |e_i|$. Put $b = s_8 \bmod 16$. The outer event is

$$S < \min(b, 16 - b). \tag{7}$$

The condition itself forces $S \leq 7$, so no separate norm cutoff is needed. A signed residue with $|e_i| \leq 7$ has a unique representative modulo sixteen.

Theorem 5.1 (Supported outer paths)**.** *On event* (7)*, a public graph made from 256 aligned 16-point rows contains all fifteen internal edges of each of the sixteen true middle paths of K_8. Enumerating graph paths, decoding them by Theorem 4.1, and applying the common carry test below retains the true numerical K_8. The event has exact probability*

$$p_{\text{out}} = \frac{48639}{16^7} = 0.00018119439482688904 \tag{8}$$

for every fixed master key.

Proof. Lemma 2.3 says that each row contains two adjacent outer quotient labels with combined count at least $16 - S \geq 9$. Project a top-twelve-bit label modulo 256, and join any two distinct observed labels in the same output lane whose counts sum to at least nine. This can add false edges, but never removes either endpoint of a true internal edge. For the 256 successive aligned input rows, the nominal middle-byte label before the final addition runs once through all 256 values: each prior rotor's middle quotient is a permutation, and each aligned quotient translation is also a permutation. The true internal edges therefore all appear. Edges crossing output-lane boundaries are unneeded for the sixteen paths.

The implementation first enumerates paths in a lane with the fewest observed edges. A completed path determines its unique core (u, v, w, t). Testing its sixteen input XORs determines each possible common external input/output XOR pair. For every other lane, equation (4) then leaves only the 16^2 choices of (u, w) with the fixed common (v, t). Test their fifteen edges and require distinct u's. This finite completion includes the true paths, even when the graph has extra edges; it imposes no path-count limit. Applying Theorem 4.1 to every completed path set includes the true raw rotor key.

For the numerical row test, invert a candidate K_8 on each of the 256 rows. For a proposed last residue b, count inverse images in the disjoint intervals $[16v + b, 16v + b + 15]$. Keep b if every row has at least

$$17 - \min(b, 16 - b) \tag{9}$$

images in one such interval. The same b must work for all rows. On the event, $S \leq \min(b, 16-b)-1$, so Lemma 2.3 gives at least $16 - S$ images in the true interval, which meets (9). All keys passing this exact necessary test are retained.

For the count, let $a_d(s)$ be the number of d-dimensional signed residue vectors of absolute norm s. Then

$$a_d(0) = 1, \qquad a_d(s) = \sum_{j=1}^{\min(d,s)} 2^j \binom{d}{j} \binom{s-1}{j-1} \quad (s > 0).$$

Choose the j nonzero coordinates, their signs, and a positive composition of s. For a fixed $s \leq 7$, exactly $15 - 2s$ values of b obey (7). Hence

$$\sum_{s=0}^{7} a_6(s)(15 - 2s) = 15 + 156 + 792 + 2628 + 6384 + 11820 + 16008 + 10836 = 48639.$$

Lemma 2.2 divides this count by 16^7. □

The public screen samples 31 consecutive words at each of twelve fixed input positions. Every candidate alignment has a complete row in a 31-word strip, and on the event its two supported labels sum to at least nine. Thus any one strip preserves its alignment. The twelve strips only reject more IVs before the 4111-word partial codebook is requested. That codebook contains all 256 rows for every alignment. The screen does not inspect a key or a hidden state.

6 One inward factorization

Write each state word $s_i = b_i + 256h_i$, $0 \le b_i, h_i < 256$, and let $h_9 = 0$. Define the translated prefix

$$C_t(p) = R_t(p) + 256h_{t+1}.$$

It obeys the same step at every stage:

$$C_{t-1}(p) = \mathcal{D}_t(C_t(p) - 256h_{t+1}) - b_t. \tag{10}$$

Indeed, substitution leaves $R_{t-1}(p) + 256h_t$. The pivot's output translation only shifts quotient labels:

$$\left\lfloor \frac{C_t(p)}{16} \right\rfloor = \left\lfloor \frac{R_t(p)}{16} \right\rfloor + 16h_{t+1} \pmod{4096}.$$

It preserves the low-nibble block partition used above.

For each $t = 6, 5, 4, 3$, define the inward event E_t: all low-nibble residues $s_2, \dots, s_6$ vanish except $s_t \bmod 16 = e_t$, where e_t is signed with $1 \le |e_t| \le 3$, and

$$2|e_t| < b_7 < 16 - 2|e_t|, \qquad b_7 = s_7 \bmod 16. \tag{11}$$

Here s_1 is handled by an unknown row alignment. For each t, the choices of (e_t, b_7) number $2(11 + 7 + 3) = 42$, so

$$q = \Pr(E_t) = 42/16^6. \tag{12}$$

The four events are disjoint because their sole nonzero residue is in a different state word.

Under any E_t, Lemma 2.3 exposes a translated stage-seven quotient cycle. Trying the sixteen possible low output residues and row alignments, the complete-cycle decoder of Section 4 includes K_7 and its output translation s_8. The implementation first checks 256 partial rows, then asks for the full first-word codebook only when a compatible numerical candidate and one next-stage graph remain. Every test is a necessary condition on the sufficient event. The full row test keeps every compatible state residue; it does not retain just the best one. The saved public context is

$$Y(p) = \mathcal{D}_7(\mathcal{D}_8(C(p)) - s_8) = R_6(p) + s_7, \tag{13}$$

together with its IV, alignment, and admissible low residues of s_7.

For E_6, inverting the recovered K_7 exposes the translated stage-six cycle as well. The exact cycle decoder and full row test give seed pairs (K_7, K_6). Numerical suffixes obtained from one IV are pooled across all saved contexts. The true suffix can therefore be extended using a different IV for each inward event. All state offsets stay with the IV from which they were obtained.

6.1 The partition join

Let $q_j(x) = \mathcal{E}_j(16x) \gg 4$ be the 4096-point quotient of rotor j, and let $\mathrm{T}_a(x) = x + a \pmod{4096}$. Suppose $K_{t+1}, \dots, K_6$ are known and put $d = 6 - t$, so $d \in \{1, 2, 3\}$. Under E_t, the quotient walk of a compatible context has the form

$$W = \mathrm{T}_{\eta_d}\, q_{t+d}\, \mathrm{T}_{\eta_{d-1}}\, q_{t+d-1} \cdots \mathrm{T}_{\eta_1}\, q_{t+1}\, \mathrm{T}_{\eta_0}\, q_t, \tag{14}$$

up to a cyclic input origin and direction. The η_j are the state quotient translations carried by (10); all are enumerated. The only nonaligned low-nibble addition is the small e_t, which creates the two-label row edges. Every known quotient and translation in (14) permutes the middle byte.

For $0 \le j \le d$, put $m_j(x) = q_{t+j}(x) \bmod 256$ on $0 \le x < 256$. Lemma 2.1 makes this the complete middle-byte quotient, independent of the input top nibble. Let $\mathcal{P}$ partition $\{0, \dots, 255\}$ into its sixteen consecutive 16-point blocks. For any 256-point permutation f, $f(\mathcal{P})$ means the unordered set of image blocks. Write m_j for the middle-byte quotient of the corresponding q_j. The exact join uses the identity

$$\left(\mathrm{T}_{-z}\, m_d^{-1}\, \mathrm{T}_{-h}\, W\, \mathrm{T}_a\right)(\mathcal{P}) = \left(\mathrm{T}_{16H}\, m_{d-1}\, \mathrm{T}_{\theta_{d-2}} \cdots m_1\, \mathrm{T}_f\right)(\mathcal{P}). \tag{15}$$

All maps in this equation are reduced modulo 256. For $d = 1$, the right side is simply $\mathcal{P}$. Here $h \in \{0, \dots, 255\}$ is the outer output translation, $z, a, f, H \in \{0, \dots, 15\}$, and the $\theta_j \in \{0, \dots, 255\}$ are inner translations that exist only at greater depth. The left side has $256 \cdot 16 \cdot 16 = 65536$ choices; the right has respectively $1, 256, 65536$ choices for depths $1, 2, 3$.

Lemma 6.1 (Exact join)**.** *On the true branch of any E_t, one pair of entries in* (15) *matches. The implemented join enumerates every matching pair, including $f = 0$. For each match, its middle-path and top-coordinate tests emit every compatible numerical K_t and quotient translation.*

Proof. Choose h, z, a to cancel the true outer translation and align the input origin with a block boundary. Undo the known outer m_d. Decompose the next translation as $z + 16H$ and the translation next to m_t as $f + 16F$. Translations by multiples of sixteen preserve $\mathcal{P}$; m_t preserves $\mathcal{P}$ by Lemma 2.1. Moving those two factors inside the image of $\mathcal{P}$ gives exactly the right side of (15); the intervening 256-point translations are the θ_j. This argument does not require $f \neq 0$.

The code encodes a partition by naming its sixteen blocks in order of first occurrence among output positions, then packs the 256 names into 128 bytes. Two codes are equal exactly when the unordered partitions are equal. It sorts the right family and compares full encodings, not hashes; every matching pair is processed. Undoing its known factors leaves the sixteen paths of an unknown middle map, with a common output-lane translation. Theorem 4.1 enumerates every matching numerical key.

It remains to restore the top quotient nibble. Write $q_j(256A + x) = 256P_{j,x}(A) + m_j(x)$, with $0 \le A < 16$. Once the middle trajectory is fixed, the carries at all known translations and the lane x of each known rotor are fixed. On the sixteen inputs with one chosen middle coordinate, the top trajectory is a composition of known nibble permutations $P_{j,x}$ and the remaining unknown top-nibble additions. Split this composition in two. Enumerate the at most 256 offset choices on one side and the at most sixteen on the other; compare their complete 16-entry maps up to cyclic input and output translations. For every equality, enumerate the sixteen input origins and compare all 4096 entries of (14). The true offsets give an equality and pass that literal comparison. No match-count assumption is used. □

The same join is called for $d = 1, 2, 3$, yielding K_5, K_4, K_3. After its quotient comparison, the code undoes the known full-word rotors and tests every aligned row against the candidate inner rotor's translated interval. On E_t there are no low-nibble perturbations before t, so Lemma 2.3 makes the true row test exact. Every surviving offset and key is emitted.

7 Initialization anchors and the first two rotors

The partition join stops at K_3. The remaining two rotors are recovered from four rounds of initialization, without an unbounded search over the other state words. Let $u_i^{(r)}$ be state word i after initialization round r, with $u_i^{(0)} = \mathrm{IV}_i$. Lemma 2.2 gives, for $i \ge 3$,

$$u_i^{(r-1)} = u_i^{(r)} - \mathcal{E}_{i-1}(u_{i-1}^{(r)}). \tag{16}$$

Known tail states and keys fill this inverse triangle. In particular, the four known tail states give

$$s_1 = \mathrm{IV}_1 + \sum_{r=1}^{4} \mathcal{E}_8(u_8^{(r)}). \tag{17}$$

For $t = 2, 3, 4$, the same triangle computes $A_t = \mathcal{E}_t(s_t)$ from $s_{t+1}, \ldots, s_{t+4}$, $K_{t+1}, \ldots, K_{t+3}$, and IV_{t+4}, without consulting K_t. To make the construction explicit, set $a_{0,j} = s_{t+1+j}$ for $0 \le j \le 3$, then

$$a_{r,j} = a_{r-1,j} - \mathcal{E}_{t+j}(a_{r-1,j-1}), \qquad 1 \le r \le j \le 3. \tag{18}$$

Induction using (16) gives $a_{r,j} = u_{t+1+j}^{(4-r)}$. Starting from $x = \mathrm{IV}_{t+4}$, replace $x \leftarrow \mathcal{D}_{t+j}(a_{j,j} - x)$ for $j = 3, 2, 1$. The result is the previous-round value feeding s_{t+1}, so $A_t = s_{t+1} - x$.

Once $K_3, s_4, \ldots, s_8$ are known on a true branch, (17) and A_3 give s_1 and $s_3 = \mathcal{D}_3(A_3)$. The triangle gives A_2. Undo the known rotors to obtain the complete public table

$$R_2(p) = \mathcal{E}_2(\mathcal{E}_1(p + s_1) + s_2), \qquad A_2 = \mathcal{E}_2(s_2). \tag{19}$$

The final sufficient event also requires $s_2 \bmod 256 \neq 0$, with independent conditional probability $15/16$.

Lemma 7.1 (Anchored pair retention)**.** *Given the correct R_2 table, s_1, A_2, and $s_2 \bmod 256 \neq 0$, the finite pair decoder emits the true (K_1, K_2, s_2). It then checks the complete word table and the literal IV initialization.*

Proof. Align the 256-point input high-byte rows by s_1. The first rotor maps each row to one high-byte class. Adding a nonzero low byte of s_2 splits it between two adjacent classes, and the second rotor permutes those classes. Thus every row has exactly two high-byte labels. The resulting 256 edges trace the high-byte quotient cycle of $\mathcal{E}_2$, with the common split count giving $s_2 \bmod 256$.

The unique preimage of A_2 in (19) is the row where the first rotor output is zero. It anchors the cycle origin and orientation. The aligned row labels then give the first rotor's complete high-byte quotient, while the anchored cycle gives the second rotor's high-byte quotient up to one of 256 input origins. The decoder enumerates every such origin. For each proposed quotient, it enumerates the raw-key lifts and checks its complete 256-entry truth table; the true raw key is among them by construction.

For clarity, the finite high-byte decoder uses the autonomous Q of Lemma 2.1 and two top-coordinate rows P_b. Put $p_0(b) = S_2(b \oplus \beta_0)$ and $p_j(b) = S_2(p_{j-1}(b) \oplus \beta_j)$ for $j = 1, 2, 3$. Direct substitution gives

$$Q(b) = S_2(p_3(b) \oplus \beta_0 \oplus \beta_1) \oplus \beta_2 \oplus \beta_3.$$

Its 65536 possible 16-entry tables are distinct. The top rows are XOR translates of

$$J_P^{u,v,w,t}(x) = S_1(S_3(S_1(S_3(S_1(x) \oplus u) \oplus v) \oplus w) \oplus t),$$

with

$$(u, v, w, t) = (\gamma_1, \alpha_2 \oplus p_1(b), \gamma_3, \alpha_0 \oplus \alpha_1 \oplus p_3(b)).$$

Exhaustively canonicalizing these 65536 nibble tables under input/output XOR gives 65535 orbits: one two-core collision and three self-stabilizing cores. The five exceptional cores have distinct (u, w) pairs. The two rows $b = 0, 1$ share (u, w) but have different v, since p_1 is a permutation. Thus at least one row is nonexceptional and fixes its core and external XORs. The raw schedule then leaves at most sixteen low-nibble choices, all scanned. These finite facts are reproduced by evaluating the printed 16-entry formulas over their complete parameter ranges and sorting the resulting packed truth tables.

For the first rotor there are at most sixteen such raw lifts; for the second there are at most $256 \cdot 16 = 4096$ origin/lift pairs. The decoder checks each pair against all 65536 entries of (19), then checks the eight initialized state words against the supplied IV. The true pair passes every comparison. □

8 True-key survival

The attack has one induction. Acquire outer IVs until the fixed budget has been examined; retain every outer numerical key from Section 5. For each such key, acquire the fixed inward budget, save every stage-seven context, and pool every seed pair. Apply the same partition join at depths one, two, and three. At each leaf, use the initialization anchors and anchored pair decoder, then replay every complete key against every saved public codebook. No score, candidate cap, or choice of a single state branch occurs in this induction.

Theorem 8.1 (True-key coverage)**.** *For every fixed 256-bit master key, use m independent uniform outer IVs and N independent uniform inward IVs for each outer candidate. The probability that the final list omits the true master key is at most*

$$\epsilon(m, N) = \left(1 - \frac{48639}{16^7}\right)^m + 3\left(1 - \frac{42}{16^6}\right)^N + \left(1 - \frac{630}{16^7}\right)^N. \tag{20}$$

At $m = 132000$, $N = 8000000$, this is

$$\epsilon(m, N) = 1.30716322171 \cdot 10^{-8} < 1.4020 \cdot 10^{-8}. \tag{21}$$

The bound concerns inclusion of the true key in the returned list.

Proof. By Theorem 5.1, one outer event retains the true K_8. On that branch, E_6 retains the true seed (K_7, K_6), and E_5, E_4, E_3 extend the true numerical suffix in turn. The proof of Lemma 6.1 includes zero translations, so there is no further state condition at E_4. For E_3, Lemma 7.1 adds $s_2 \bmod 256 \neq 0$. Given E_3, its upper four bits of the low byte are unrestricted and uniform by Lemma 2.2; this adds the factor $15/16$, giving $42 \cdot 15/16^7 = 630/16^7$. The correct branch passes literal initialization and all public-ciphertext replay.

The absence probabilities for the outer event and the four inward event families are respectively $(1 - p_{\text{out}})^m$, $(1 - q)^N$ three times, and $(1 - 630/16^7)^N$. Union bound them. Independence among the different families is not assumed. False branches can increase work but cannot remove the true branch because every compatible candidate is processed. □

$$\boxed{\Pr\{\text{true master key is returned}\} \geq 1 - 1.30716322171 \cdot 10^{-8}.}$$

References

[1] A. Vahi and S. Jafarali Jassbi. SEPAR: A New Lightweight Hybrid Encryption Algorithm with a Novel Design Approach for IoT. *Wireless Personal Communications* 114, 2283–2314, 2020. doi:10.1007/s11277-020-07476-y.